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Code:

```python
import pandas as pd

# Load dataset
df = pd.read_csv("shop_data.csv")

Original_rows = len(df)

# Display Dataset
print("=== Original Dataset ===")
print(f"Shape: {df.shape}")
print("\nHead of dataset:\n", df.head())
print("\nInfo:\n")
print(df.info())
print("\nDescription:\n", df.describe())

# Check missing values
C_null = df.isnull().sum().sum()

if C_null > 0:
    # Filling missing values
    df['quantity'] = df['quantity'].fillna(0)
    df['price'] = df['price'].fillna(df['price'].median())

    C_null = df.isnull().sum().sum()

# Calculating missing values
nonNULL_count = Original_rows - C_null
Total_nullRemoved = Original_rows - nonNULL_count

# Removing missing values
df.dropna()

# Remove duplicates
print("\n=== Duplicates ===")
duplicates = df[df.duplicated()]
No_duplicates = len(duplicates)
if No_duplicates > 0:
    df = df.drop_duplicates()
    print("Duplicates removed.")

# Remove Outliers
df = df.sort_values("quantity")
Q1 = df["quantity"].quantile(0.25)
Q3 = df["quantity"].quantile(0.75)
IQR = Q3 - Q1

outliers = df[
    (df["quantity"] < (Q1 - 1.5 * IQR)) |
    (df["quantity"] > (Q3 + 1.5 * IQR))
]
No_outliers = len(outliers)

df = df.drop(outliers.index)

# Total Cleaned Rows
cleaned_rows = Original_rows - Total_nullRemoved - No_outliers - No_duplicates

# Final Report
print("\n----- DATA CLEANING REPORT -----")
print(f"\nOriginal Rows = {Original_rows}")
print(f"\nMissing Values Removed = {Total_nullRemoved}")
print(f"\nDuplicate Rows Removed = {No_duplicates}")
print(f"\nOutliers Removed = {No_outliers}")
print(f"\nFinal Cleaned Rows = {cleaned_rows}")

print("Data cleaned successfully!")

# Save the cleaned dataset
def saveCSV():
    df.to_csv('cleaned_shop_data.csv', index=False)
    print("\nCleaned dataset saved as 'cleaned_shop_data.csv'")

saveCSV()
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS\June 2026\Week 4\Tasks 29-06 - miniProject> python .\data_cleaning.py
=== Original Dataset ===
Shape: (1000, 10)

Head of dataset:
                                sale_id product_name   price  quantity   sale_date     customer_name              customer_email                                   customer_address payment_method store_location
0  e819253c-00af-4003-b1c4-64fdf19a7597       nearly  188.60         4  2023-12-31      Angela Lopez       tinaglover@example.net     7925 Mark Flat Apt. 755\nPort Robert, CO 42291         PayPal    Danielshire
1  b125778e-1944-4a35-90c3-5c8f864e6326        hotel   55.93         3  2024-04-05        John Terry        rebecca14@example.net             470 Jacob Greens\nSmithland, FM 46143         PayPal     Monicaside
2  2705089b-0b13-41c8-b38a-698c6edaa270        range  253.77         3  2024-03-11      David Norton        tlawrence@example.org        57925 Morgan Unions\nDebbieville, DC 69607    Credit Card       Kentberg
3  121bc696-0f16-44a0-86f3-c1433552eed3        raise  326.35         6  2023-12-07  Alejandro Austin  williamsdouglas@example.org  50394 Michael Fork Suite 157\nNew Brandonton, ...         PayPal  New Christian
4  bb8e1780-cb8b-4ed4-a53f-18cc165d46c7       former  389.81         1  2023-02-19  Daniel Rodriguez   andradeantonio@example.net                         USS Hamilton\nFPO AE 31717         PayPal  Edwardchester

Info:

<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   sale_id           1000 non-null   str
 1   product_name      1000 non-null   str
 2   price             1000 non-null   float64
 3   quantity          1000 non-null   int64
 4   sale_date         1000 non-null   str
 5   customer_name     1000 non-null   str
 6   customer_email    1000 non-null   str
 7   customer_address  1000 non-null   str
 8   payment_method    1000 non-null   str
 9   store_location    1000 non-null   str
dtypes: float64(1), int64(1), str(8)
memory usage: 78.3 KB
None

Description:
             price     quantity
count  1000.000000  1000.000000
mean    252.332360     5.338000
std     144.492106     2.846025
min       5.310000     1.000000
25%     128.147500     3.000000
50%     250.405000     5.000000
75%     378.737500     8.000000
max     499.710000    10.000000

=== Duplicates ===

----- DATA CLEANING REPORT -----

Original Rows = 1000

Missing Values Removed = 0

Duplicate Rows Removed = 0

Outliers Removed = 0

Final Cleaned Rows = 1000
Data cleaned successfully!

Cleaned dataset saved as 'cleaned_shop_data.csv'
```